Algebraic Two-Electron Integrals on the Prolate Spheroidal Lattice*

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Paper 11 established the prolate spheroidal lattice as the natural geometry for diatomic molecules, solving H_2^+ to machine precision (spectral Laguerre solver) with zero free parameters. The extension to two electrons requires electron-electron repulsion integrals $\langle\phi_i|1/r_{12}|\phi_j\rangle$ over the six-dimensional two-particle configuration space. The standard approach—numerical quadrature on a grid in prolate spheroidal coordinates—converges slowly due to the Coulomb singularity at $r_{12} = 0$ and saturates at ∼80% of the exact H_2 dissociation energy D_e regardless of basis size.

We replace numerical integration with algebraic evaluation using the Neumann expansion of $1/r_{12}$ in prolate spheroidal coordinates, which decomposes each six-dimensional integral into a finite sum of products of one-dimensional auxiliary integrals computable by recurrence relations. The resulting two-electron integrals are exact within the Neumann truncation order L, with no six-dimensional quadrature and no fitted parameters. The one step historically left to numerical integration—the log-singular second-kind Legendre moment B_l—also has a closed form (Sec. III D): the only transcendental content is the exponential integral E_1, the Euler–Mascheroni constant γ and ln, so *in the σ sector the reduction is fully quadrature-free.* (The $|m| \geq 1$ channels keep the same kernel; generalising the same A_l, B_l, X_l recurrences to associated Legendre functions (Sec. VII) gives a moment-recurrence engine that is *stable* in the δ ($|m| = 2$) sector where the differentiation route diverges, and—with the same closed-form seeds—is likewise fully quadrature-free.)

A systematic convergence study on H_2 at $R = 1.4011$ bohr demonstrates that the Neumann algebraic V_{ee} yields 92.2% of the exact D_e with 27 basis functions, compared to 79.6% for numerical V_{ee} at the same basis size—a uniform 12–20 percentage point improvement across all basis sizes from 1 to 72 functions. The Neumann result plateaus at 92.4%, confirming that V_{ee} integration error has been eliminated. [**MEASURED**] The remaining 7.6% gap is a basis-space limit, and we identify which one. The basis used here is φ-independent, so it spans only $m_1 = m_2 = 0$ configurations; but a $^1\Sigma_g^+$ state constrains the *total* $M = m_1 + m_2$ to zero, not each m_i separately, so the π^2, δ^2, ... configurations—every one of them $^1\Sigma_g^+$—are absent by construction. Restoring them in the same basis with the same Neumann kernel gives 99.1% of D_e at $|m| \leq 1$ (stability envelope 99.0–99.1%; Sec. VII), a gain of 11.6 mHa that no growth along the σ axis reproduces (near-tripling the σ basis is worth 0.34 mHa). [**MEASURED**] That 99.1% is where the *monomial* radial basis ξ^j stalls—a Hankel moment problem whose overlap matrix is past double precision, not a structural ceiling. Rebasing the same span onto an orthogonal-polynomial family is an exact change of basis that removes the linear dependence, and the energy then climbs monotonically and variationally to 99.81% of D_e (0.32 mHa, inside chemical accuracy) at a larger truncation and the variationally optimal exponent (Sec. VII A). [**WITHDRAWN**] This paper previously attributed the gap to the electron-electron cusp requiring non-analytic $r_{12}^{1/2}$ and $r_{12} \ln r_{12}$ terms, and read that as motivation for hyperspherical coordinates. Neither holds: no such term appears in the calculation above, nor in the grid-based prolate spheroidal FE-DVR treatment of Tao, McCurdy and Rescigno, an accurate two-electron H_2 calculation in these same coordinates. [**MEASURED**] An explicit-r_{12} (James–Coolidge) extension of the algebraic engine, in which every r_{12} matrix element reduces to the same $A\,K_0 - B\,K_1$ Neumann kernels as V_{ee} (no quadrature), reaches 0.053 mHa above the exact value—a sixfold reduction of the CI residual—confirming constructively that ordinary integer powers of r_{12}, not non-analytic terms, close the gap (Sec. VII B). The same reduction—on one centre the classical Hylleraas-CI treatment of many-electron correlated integrals—extends without a resolution of identity to a heteronuclear diatomic (HeH^+) and through the *complete* many-body matrix elements of a two-center four-electron explicit-r_{12} CI: the genuinely four-body Coulomb bridge and the two-center three-body triangle are both reduced RI-free—the latter through a low-rank expansion of its non-separable two-point kernel contracted mode-by-mode with the general-m prolate Neumann Coulomb—so the entire 2×2 CI energy assembles analytically, matching a same-geminal variational Monte-Carlo reference to sub-mHa and lying above the exact value (variational) (Sec. VII B).

I. INTRODUCTION

The Geometric Vacuum program identifies and exploits the *natural geometry* of each quantum system—the coordinate system where the Schrödinger equation

* Paper 12 in the Geometric Vacuum series

separates—replacing continuous integration with algebraic operations on quantum number labels [1–3]. For atoms, the natural geometry is Fock's three-sphere S^3 [8], where the Coulomb potential disappears into the conformal metric and the Schrödinger equation becomes a pure Laplace–Beltrami eigenvalue problem. For diatomic molecules, the natural geometry is the prolate spheroid, where the two-center Coulomb problem separates into coupled ordinary differential equations [4].

Paper 11 implemented this principle for the one-electron molecule H_2^+, reaching machine precision (spectral Laguerre solver) with zero free parameters. The extension to two electrons introduces a qualitatively new challenge: the electron-electron repulsion integral

$$V_{ee}^{ij} = \left\langle \phi_i \left| \frac{1}{r_{12}} \right| \phi_j \right\rangle, \tag{1}$$

which couples the coordinates of both electrons and cannot be evaluated by separation of variables. In the prolate spheroidal Hylleraas-type CI framework, these integrals are six-dimensional (three coordinates per electron). The σ-only basis adopted below reduces the effective dimensionality to four; Sec. VI shows that this is a *restriction* on the configuration space, not a symmetry of the ${}^1\Sigma_g^+$ state, and measures what it costs.

The standard approach computes Eq. (1) numerically, using the complete elliptic integral $K(k)$ to perform the azimuthal average of $1/r_{12}$ and Gauss–Legendre quadrature for the remaining four dimensions $(\xi_1, \eta_1, \xi_2, \eta_2)$. This approach has two fundamental limitations:

1. The Coulomb singularity at $r_{12} = 0$ (the electron-electron coalescence point) is poorly resolved by polynomial quadrature, requiring increasingly fine grids for convergence.

2. The exponential decay $e^{-\alpha(\xi_1+\xi_2)}$ of the basis functions extends to $\xi \to \infty$, requiring truncation of the semi-infinite radial domain.

As we demonstrate in Section V, these limitations cause the numerical V_{ee} to saturate at ∼80% of the exact H_2 dissociation energy, regardless of how many basis functions are employed.

This paper replaces numerical quadrature with the Neumann expansion of $1/r_{12}$ in prolate spheroidal coordinates [9–11], which converts each V_{ee} matrix element into a finite sum of products of one-dimensional integrals. These auxiliary integrals—angular moments $C_l(p)$, exponential moments $A_n(\alpha)$, and first- and second-kind Legendre moments $A_l(p, \alpha)$ and $B_l(p, \alpha)$—are computed by recurrence relations and closed forms. The log-singular second-kind moment B_l, once the sole step left to adaptive quadrature, is now itself closed-form (Sec. III D): a single log-moment against Q_0 whose primitive reduces to E_1, γ and ln. The six-dimensional repulsion integral is thereby reduced, *in the σ sector*, to a finite sum of algebraic operations on quantum number labels with no numerical integration of any dimension; Sec. VII states what the $|m| \geq 1$ extension costs.

The structure of this paper is as follows. Section II derives the Neumann expansion and its factorization for σ-state matrix elements. Section III defines the auxiliary integrals and their recurrence relations. Section IV describes the computational pipeline. Section V presents the convergence study comparing numerical and algebraic V_{ee}. Section VI diagnoses the origin of the remaining 7.6% gap and measures its removal. Section VIII discusses implications for the GeoVac program. Section IX concludes.

II. THE NEUMANN EXPANSION

A. Expansion of $1/r_{12}$ in prolate spheroidal coordinates

The Coulomb kernel $1/|\mathbf{r}_1 - \mathbf{r}_2|$ can be expanded in prolate spheroidal coordinates (ξ, η, φ) using the Neumann expansion [9, 12]:

$$\frac{1}{r_{12}} = \frac{2}{R} \sum_{l=0}^{\infty} \sum_{m=0}^{l} (2-\delta_{m0})\,(-1)^m\,(2l+1) \left[\frac{(l-m)!}{(l+m)!}\right]^2 P_l^m(\xi_<)\, Q_l^m(\xi_>) \tag{2}$$

where $\xi_< = \min(\xi_1, \xi_2)$, $\xi_> = \max(\xi_1, \xi_2)$, P_l^m and Q_l^m are associated Legendre functions of the first and second kind, and R is the internuclear distance. **[MEASURED]** Earlier versions of this paper printed Eq. (2) without the $(-1)^m$ and $(2l+1)$ factors and with the factorial ratio unsquared. That form does not reproduce $1/r_{12}$—checked pointwise, it diverges—whereas the expression above agrees with $1/|\mathbf{r}_1 - \mathbf{r}_2|$ to 2×10^{-6} by $l = 10$ at well-separated ξ. The error went undetected because every calculation in *earlier versions* of this paper used only the $m = 0$ specialisation Eq. (3), which was derived independently—from the Legendre completeness relation, as stated below—rather than specialised from the printed general form. Two of the three discrepancies do vanish at $m = 0$; the $(2l + 1)$ does not, and Eq. (3) carries it because that independent derivation supplies it. Sec. VII is the first use of the general form, which is why the error surfaced only now.

The basis of Eq. (4) below carries no φ dependence, so for *that basis* the azimuthal integration $\int_0^{2\pi} d\varphi_1 \int_0^{2\pi} d\varphi_2$ retains only the $m = 0$ component:

$$\left\langle \frac{1}{r_{12}} \right\rangle_{\varphi} = \frac{8\pi^2}{R} \sum_{l=0}^{\infty} (2l+1)\, P_l(\xi_<)\, Q_l(\xi_>)\, P_l(\eta_1)\, P_l(\eta_2), \tag{3}$$

where $P_l \equiv P_l^0$ and $Q_l \equiv Q_l^0$ are ordinary Legendre functions. The factor $8\pi^2 = 2 \times 4\pi^2$ combines the $2/R$ prefactor of the Neumann expansion (2) with the $4\pi^2$ from $\int_0^{2\pi}\int_0^{2\pi} \cos 0\, d\varphi_1\, d\varphi_2$. The $(2l + 1)$ factor arises from the Legendre polynomial normalization in the angular completeness relation $\int_{-1}^{1} P_l(\eta)\, P_{l'}(\eta)\, d\eta = 2\delta_{ll'}/(2l + 1)$.

Equation (3) is the key identity for the σ-only sector: it separates the two-electron Coulomb kernel into a sum of products of functions of individual coordinates. **[SCOPE]** It is *not* the whole kernel for a ${}^1\Sigma_g^+$ state. Retaining only $m = 0$ is a restriction to σ configurations, not a consequence of Σ symmetry: the Σ label fixes $M = m_1 + m_2 = 0$, which π^2 and δ^2 configurations satisfy with $m_2 = -m_1 \neq 0$. The terms with $m \neq 0$ in Eq. (2) are precisely what couples those configurations to the σ sector. Sections VI and VII measure the cost of dropping them. Each term in the sum involves $P_l(\xi_<)Q_l(\xi_>)$, which couples the two radial coordinates, and $P_l(\eta_1)P_l(\eta_2)$, which factorizes completely.

B. Matrix element factorization

Consider two basis functions of the Hylleraas type in prolate spheroidal coordinates:

$$\phi_i(\xi_1, \eta_1, \xi_2, \eta_2) = \mathcal{N}_i \, e^{-\alpha(\xi_1+\xi_2)} \, \xi_1^{j_i} \xi_2^{k_i} \, \eta_1^{l_i} \eta_2^{m_i} + (1 \leftrightarrow 2), \tag{4}$$

where the $(1 \leftrightarrow 2)$ exchange ensures bosonic spatial symmetry for the singlet state, and (j_i, k_i, l_i, m_i) are nonnegative integer quantum numbers. The exponential parameter α controls the spatial extent; for all basis functions in a given calculation we use a common α optimized variationally.

The V_{ee} matrix element between ϕ_i and ϕ_j is

$$V_{ee}^{ij} = \iint \phi_i^* \, \frac{1}{r_{12}} \, \phi_j \; J(\xi_1, \eta_1) \, J(\xi_2, \eta_2) \, d\xi_1 \, d\eta_1 \, d\xi_2 \, d\eta_2, \tag{5}$$

where $J(\xi, \eta) = (R/2)^3(\xi^2 - \eta^2)$ is the prolate spheroidal Jacobian (with the 2π azimuthal factors already absorbed into Eq. (3)).

Substituting the Neumann expansion (3) and expanding the product of two basis functions, each unsymmetrized term contributes a product of the form

$$e^{-2\alpha(\xi_1+\xi_2)} \, \xi_1^{P_1} \xi_2^{P_2} \, \eta_1^{Q_1} \eta_2^{Q_2}, \tag{6}$$

where $P_1 = j_i + j_j$, $P_2 = k_i + k_j$, $Q_1 = l_i + l_j$, $Q_2 = m_i + m_j$ are the combined powers from bra and ket.

The Jacobian factor $(\xi^2 - \eta^2)$ for each electron expands the integral into four sub-terms with shifted powers:

$$\begin{aligned}
&(+1): \; (P_1{+}2, \; P_2{+}2, \; Q_1, \; \; Q_2), \\
&(-1): \; (P_1{+}2, \; P_2, \; \; \; Q_1, \; \; Q_2{+}2), \\
&(-1): \; (P_1, \; \; \; P_2{+}2, \; Q_1{+}2, \; Q_2), \\
&(+1): \; (P_1, \; \; \; P_2, \; \; \; Q_1{+}2, \; Q_2{+}2).
\end{aligned} \tag{7}$$

For each such sub-term with effective powers $(\tilde{P}_1, \tilde{P}_2, \tilde{Q}_1, \tilde{Q}_2)$, the matrix element factorizes as

$$\frac{\pi^2 R^5}{8} \sum_{l=0}^{L} (2l+1) \, X_l(\tilde{P}_1, \tilde{P}_2, 2\alpha) \, C_l(\tilde{Q}_1) \, C_l(\tilde{Q}_2), \tag{8}$$

where $C_l(q) = \int_{-1}^{+1} \eta^q \, P_l(\eta) \, d\eta$ are angular moment integrals, and

$$X_l(p_1, p_2, \alpha) = \int_1^\infty \int_1^\infty \xi_1^{p_1} \xi_2^{p_2} \, e^{-\alpha(\xi_1+\xi_2)} \, P_l(\xi_<) \, Q_l(\xi_>) \, d\xi_1 \, d\xi_2 \tag{9}$$

is the two-dimensional radial integral coupling the two electrons through the $P_l(\xi_<)Q_l(\xi_>)$ kernel.

The prefactor $\pi^2 R^5/8$ arises from $(2/R) \times (R/2)^6 \times (2\pi)^2 = \pi^2 R^5/8$, combining the Neumann prefactor $2/R$, the two Jacobians $(R/2)^3$ each, and the azimuthal integration $4\pi^2$.

C. Selection rules

The angular integrals $C_l(q)$ obey a powerful selection rule:

$$C_l(q) = 0 \quad \text{if } q < l \text{ or } (q - l) \text{ is odd.} \tag{10}$$

This follows from the orthogonality of Legendre polynomials and parity: $P_l(\eta)$ has parity $(-1)^l$, so $\eta^q P_l(\eta)$ integrates to zero over $[-1, 1]$ unless $q + l$ is even, i.e., $(q - l)$ is even. The condition $q \geq l$ ensures that η^q has enough powers to produce a nonzero overlap with P_l.

For a basis with maximum η-powers $q_{\max}$, the effective Neumann truncation is $l_{\text{eff}} = \min(L, q_{\max})$, where L is the nominal truncation order. We use $L = 20$ throughout, which is more than sufficient since our basis functions have $q_{\max} \leq 8$.

III. AUXILIARY INTEGRALS AND RECURRENCE RELATIONS

The factorized form (8) reduces each V_{ee} matrix element to products of three families of one-dimensional integrals: $C_l(q)$, $A_n(\alpha)$ and its Legendre generalizations, and $B_l(p, \alpha)$. The C_l and A_n families are computed by pure recurrence relations; the log-singular second-kind moment B_l has a closed form (Sec. III D), with adaptive quadrature retained only as an independent numerical check.

A. Angular moments $C_l(q)$

$$C_l(q) = \int_{-1}^{+1} \eta^q \, P_l(\eta) \, d\eta. \tag{11}$$

The base cases are

$$C_0(q) = \begin{cases} 2/(q+1) & q \text{ even}, \\ 0 & q \text{ odd}, \end{cases} \qquad C_1(q) = \begin{cases} 0 & q \text{ even}, \\ 2/(q+2) & q \text{ odd}. \end{cases} \tag{12}$$

Higher orders follow from the Legendre recurrence:

$$(l+1)\,C_{l+1}(q) = (2l+1)\,C_l(q+1) - l\,C_{l-1}(q). \quad (13)$$

The recurrence $C_{l+1}(q)$ requires $C_l(q+1)$, so computing $C_{l_{\max}}(q)$ requires workspace up to $q + l_{\max}$.

B. Exponential moments $A_n(\alpha)$

$$A_n(\alpha) = \int_1^\infty \xi^n\, e^{-\alpha\xi}\, d\xi. \quad (14)$$

The base case is $A_0(\alpha) = e^{-\alpha}/\alpha$, and the upward recurrence is

$$A_n(\alpha) = \frac{n\,A_{n-1}(\alpha) + e^{-\alpha}}{\alpha}. \quad (15)$$

For negative indices, $A_{-1}(\alpha) = E_1(\alpha)$ (the exponential integral) and the downward recurrence

$$A_{-n-1}(\alpha) = \frac{e^{-\alpha} - \alpha\,A_{-n}(\alpha)}{n} \quad (16)$$

extends to arbitrary negative n.

C. First-kind Legendre moments $A_l(p,\alpha)$

$$A_l(p,\alpha) = \int_1^\infty \xi^p\, e^{-\alpha\xi}\, P_l(\xi)\, d\xi. \quad (17)$$

The base cases are $A_0(p,\alpha) = A_p(\alpha)$ and $A_1(p,\alpha) = A_{p+1}(\alpha)$, since $P_0(\xi) = 1$ and $P_1(\xi) = \xi$. The Legendre recurrence gives

$$(l+1)\,A_{l+1}(p,\alpha) = (2l+1)\,A_l(p+1,\alpha) - l\,A_{l-1}(p,\alpha). \quad (18)$$

D. Second-kind Legendre moments $B_l(p,\alpha)$

$$B_l(p,\alpha) = \int_1^\infty \xi^p\, e^{-\alpha\xi}\, Q_l(\xi)\, d\xi. \quad (19)$$

These integrals involve the Legendre function of the second kind $Q_l(\xi)$, which has a logarithmic singularity at $\xi = 1$:

$$Q_0(\xi) = \tfrac{1}{2}\ln\frac{\xi+1}{\xi-1}, \qquad Q_l(\xi) \sim -\tfrac{1}{2}\ln(\xi-1) + O(1) \quad \text{as } \xi \to 1^+. \quad (20)$$

The log singularity is integrable (the integrand behaves as $\ln(\xi-1)\,e^{-\alpha\xi} \to \ln(\xi-1)\,e^{-\alpha}$ near $\xi = 1$), but it defeats polynomial quadrature rules such as Gauss–Laguerre, which assume smooth integrands.

Rather than integrate (19) numerically, we reduce it to closed form. Write $Q_l = P_l\,Q_0 - W_{l-1}$, where $W_{l-1}(\xi) = \sum_{k=1}^{l} P_{k-1}(\xi)P_{l-k}(\xi)/k$ is a polynomial and $Q_0(\xi) = \frac{1}{2}\ln\frac{\xi+1}{\xi-1}$. The polynomial part contributes ordinary exponential moments $A_n(\alpha)$; the entire transcendental content is carried by one family of *log-moments* against Q_0,

$$B_l(p,\alpha) = \sum_n \left(\xi^p P_l\right)_n L_n(\alpha) - \sum_n \left(\xi^p W_{l-1}\right)_n A_n(\alpha), \qquad L_n(\alpha) = \quad (21)$$

where $(\,\cdot\,)_n$ denotes the coefficient of ξ^n. The primitive L_n is itself closed-form. Splitting $Q_0 = \frac{1}{2}[\ln(\xi+1) - \ln(\xi-1)]$ and substituting $u = \xi \mp 1$,

$$L_n(\alpha) = \tfrac{1}{2}\left(L_n^+ - L_n^-\right),$$
$$L_n^- = e^{-\alpha}\sum_{j=0}^{n}\binom{n}{j}\frac{j!\,(H_j - \gamma - \ln\alpha)}{\alpha^{\,j+1}}, \quad (22)$$
$$L_0^+ = \frac{e^{-\alpha}\ln 2}{\alpha} + \frac{e^{\alpha}}{\alpha}\,E_1(2\alpha), \qquad L_n^+ = \frac{e^{-\alpha}\ln 2}{\alpha} + \frac{n}{\alpha}L_{n-1}^+ + \frac{1}{\alpha}\Big[\sum \quad (23)$$

with H_j the harmonic number, γ the Euler–Mascheroni constant, and E_1 the exponential integral. The whole B_l table then follows from the $l = 0, 1$ seeds by the Legendre recurrence

$$(l+1)\,B_{l+1}(p,\alpha) = (2l+1)\,B_l(p+1,\alpha) - l\,B_{l-1}(p,\alpha). \quad (24)$$

Thus the log-singular moment introduces *no* numerical integration: its minimal transcendental content is exactly $\{E_1(2\alpha),\,\gamma,\,\ln\}$, the exponential-integral signature of the Coulomb / Q_0 projection. Adaptive quadrature [20] with subdivision near $\xi = 1$ is retained only as an independent numerical check, agreeing with Eqs. (21)–(23) to $\lesssim 10^{-20}$ (backing: `tests/test_paper12_general_m_neumann.py`).

E. Two-dimensional radial integral $X_l(p_1,p_2,\alpha)$

The integral (9) couples the two radial coordinates through the $P_l(\xi_<)Q_l(\xi_>)$ kernel. We split it into two ordered regions:

$$X_l(p_1,p_2,\alpha) = \int_1^\infty\!\int_1^{\xi_2} \xi_1^{p_1}\xi_2^{p_2}\, e^{-\alpha(\xi_1+\xi_2)} P_l(\xi_1)\,Q_l(\xi_2)\,d\xi_1\,d\xi_2 + (1 \leftrightarrow 2). \quad (25)$$

The inner integral $S_l(p,\alpha,\xi_0) = \int_1^{\xi_0} \xi^p\, e^{-\alpha\xi}\, P_l(\xi)\,d\xi$ is the *incomplete* first-kind Legendre moment. Expanding $\xi^p P_l(\xi) = \sum_j a_j\,\xi^j$ as a polynomial and integrating each

monomial by parts against $e^{-\alpha\xi}$:

$$S_l(p,\alpha,\xi_0) = A_l(p,\alpha) - e^{-\alpha\xi_0} \sum_j a_j \sum_{k=0}^{j} \frac{j!}{(j-k)!\,\alpha^{k+1}}\,\xi_0^{j-k} + (\text{boundary at } \xi=1). \tag{26}$$

Substituting into the outer integral and combining the $e^{-\alpha\xi_0}$ factor from S_l with the $e^{-\alpha\xi_2}$ factor from the outer integrand yields $e^{-2\alpha\xi_2}$ terms, which are evaluated as $B_l(\cdot, 2\alpha)$:

$$X_l = A_l(p_1)\,B_l(p_2) + A_l(p_2)\,B_l(p_1) - (\text{correction terms involving } B_l(\cdot,2\alpha)). \tag{27}$$

The full expression involves sums over the polynomial coefficients of $\xi^{p_i} P_l(\xi)$ and is implemented as described in Section IV.

IV. IMPLEMENTATION

A. The algebraic V_{ee} pipeline

The computation of the V_{ee} matrix proceeds in four stages:

1. **Table precomputation.** For a given exponential parameter α and basis quantum numbers, compute:
   - The angular moment table $C_l(q)$ for $l \le l_{\text{eff}}$, $q \le 2q_{\max} + 2$, via the recurrence (13). Cost: $O(l_{\max} \cdot q_{\max})$, negligible.
   - The first-kind Legendre moments $A_l(p, 2\alpha)$ via the recurrence (18). Cost: $O(l_{\max} \cdot p_{\max})$.
   - The second-kind Legendre moments $B_l(p, 2\alpha)$ and $B_l(p, 4\alpha)$ in closed form (Eqs. (21)–(24)). Cost: $O(l_{\max} \cdot p_{\max})$ arithmetic, no quadrature; this replaced the former adaptive-quadrature seeding, which had dominated the V_{ee} build ($\sim$40$\times$ faster, identical values).
2. X_l **table.** Compute $X_l(p_1, p_2, 2\alpha)$ for all $l \le l_{\text{eff}}$, $p_1, p_2 \le p_{\max} + l_{\max} + 2$, using Eq. (27) with the precomputed A_l and B_l tables. Cost: $O(l_{\max} \cdot p_{\max}^2 \cdot l_{\max})$ arithmetic operations.
3. **Matrix assembly.** For each pair (i, j) of basis functions, expand the bra$\times$ket product into unsymmetrized terms, apply the Jacobian expansion (7), and sum the Neumann series (8) using precomputed C_l and X_l tables. Cost per element: $O(l_{\max})$ (a single loop over l with table lookups).
4. **Symmetrization.** Exploit $V_{ee}^{ij} = V_{ee}^{ji}$ to compute only the upper triangle.

The total cost scales as $O(N_{\text{bf}}^2 \cdot l_{\max})$ for the matrix assembly, plus $O(l_{\max}^2 \cdot p_{\max}^2)$ for table precomputation. For $N_{\text{bf}} = 27$ and $l_{\max} = 8$ (the effective truncation), the V_{ee} matrix is computed in $\sim$1 s.

B. Hybrid Hamiltonian: Numba $T + V_{ne}$ plus algebraic V_{ee}

The one-electron integrals (kinetic energy T and nuclear attraction V_{ne}) are computed analytically using integration by parts for the kinetic energy derivatives and direct quadrature for V_{ne}, accelerated by Numba JIT compilation [19]. A dedicated Numba kernel `hamiltonian_tvne_p0_kernel` computes $T+V_{ne}$ without V_{ee}, avoiding the expensive elliptic integral computation that dominates the full numerical kernel.

The complete Hamiltonian is assembled as

$$H = H_{T+V_{ne}}^{(\text{Numba})} + V_{ee}^{(\text{Neumann})}, \tag{28}$$

where the first term uses grid-based quadrature (for the non-singular T and V_{ne} integrals) and the second uses the algebraic Neumann expansion (for the singular V_{ee} integrals). This hybrid approach plays to the strengths of each method: grids handle smooth integrands efficiently, while the Neumann expansion handles the Coulomb singularity exactly.

C. Computational cost comparison

For the V_{ee} matrix alone:

- **Numerical** (elliptic integral quadrature): $O(N_{\text{bf}}^2 \cdot N_\xi^2 \cdot N_\eta^2)$. At $N_\xi = 30$, $N_\eta = 20$: $\sim$3 s for $N_{\text{bf}} = 6$, scaling as $N_\xi^2 N_\eta^2$ per element.
- **Algebraic** (Neumann expansion): $O(N_{\text{bf}}^2 \cdot l_{\max} + l_{\max}^2 \cdot p_{\max}^2)$, pure arithmetic throughout—the closed-form B_l tables (Sec. III D) removed the former quadrature entirely.

The algebraic approach is both faster (the closed-form B_l made the V_{ee} build $\sim$40$\times$ faster than the earlier quadrature-seeded path) and *exact within the Neumann truncation*, whereas the numerical approach always carries grid truncation error.

V. CONVERGENCE STUDY

A. Computational setup

We compute the H_2 ground-state energy at $R = 1.4011$ bohr (near-equilibrium) using a Hylleraas-type CI with basis functions (4). The basis is parameterized by $(j_{\max}, l_{\max})$, which control the maximum powers of ξ and η respectively. The exponential parameter α is optimized variationally at each basis size using golden-section search over $\alpha \in [0.5, 2.5]$.

For each basis size, we compute the ground-state energy using both numerical and Neumann V_{ee}:

- **Numerical:** Full grid quadrature with elliptic integral $K(k)$ on a 30×20 grid ($N_\xi \times N_\eta$).

- **Neumann:** Algebraic V_{ee} via Eq. (8) with $L = 20$.

The exact non-relativistic H_2 energy at this geometry is $E_{\text{exact}} = -1.174475$ Ha [14], giving a dissociation energy $D_e = 0.174475$ Ha relative to two separated hydrogen atoms ($E = -1.0$ Ha).

B. Results

Table I presents the complete convergence study. The basis configurations range from a single function ($j = 0$, $l = 0$, $N = 1$) to 72 functions ($j = 3$, $l = 3$).

Three features of Table I deserve emphasis:

a. 1. Systematic improvement. The Neumann V_{ee} yields a lower (better) energy than numerical V_{ee} at *every* basis size, by 12–20 percentage points. This is not a sporadic improvement from fortuitous cancellation; it is a uniform advantage arising from the elimination of grid truncation error in V_{ee}.

b. 2. Numerical plateau at ∼80%. The numerical V_{ee} saturates at ∼80% D_e from $N = 27$ onward: adding basis functions from 27 to 72 improves the energy by only 0.5 percentage points (79.6% → 80.1%). This saturation is caused by grid truncation error in the V_{ee} integrals, which introduces a systematic positive bias in the electron-electron repulsion. The basis is flexible enough to represent a better wavefunction, but the inaccurate V_{ee} prevents it from being realized variationally.

c. 3. Neumann plateau at 92.4%. The Neumann V_{ee} also plateaus, but at a much higher level: 92.2% at $N = 27$ and 92.4% at $N = 72$. Since the Neumann V_{ee} is exact (within the $L = 20$ truncation, which is converged by the selection rule), this plateau is *not* caused by integration error. It identifies a genuine basis completeness limit, analyzed in Section VI.

C. The sweet spot: 27 basis functions

The data reveal a clear efficiency optimum at $N = 27$ ($j = 2$, $l = 2$): this configuration achieves 92.2% D_e in 705 s, whereas tripling the basis to $N = 72$ gains only 0.2 percentage points at 7.5× the computational cost. The diminishing returns beyond $N = 27$ reflect the fact that additional σ-type basis functions with higher ξ and η powers cannot access the missing physics—which is not in the σ sector at all (Sections VI and VII).

D. Grid dependence of numerical V_{ee}

To confirm that the numerical plateau is grid-limited, we performed a grid convergence study for the 6-function basis ($j = 1$, $l = 0$) at fixed $\alpha = 1.0$:

With the same 6 basis functions, D_e% increases from 66% to 85.7% as the grid is refined from 12×8 to 60×40. The convergence is slow (∼1.3 percentage points per grid doubling) and the cost grows quadratically. In contrast, the Neumann result for this basis is grid-independent at 61.8% D_e (the basis itself limits accuracy at 6 functions; the V_{ee} angular series terminates exactly, and the B_l moment is closed-form, so no quadrature enters).

VI. DIAGNOSIS OF THE REMAINING 7.6% GAP

A. What the plateau tells us

The Neumann plateau at 92.4% establishes that the V_{ee} integration is converged: the Neumann series at $L = 20$ is exact for our basis (the selection rule (10) caps the effective l well below 20). The 7.6% gap is therefore a property of the *basis itself*, not of the integral evaluation.

The basis functions (4) are products of integer powers of ξ and η times a common exponential $e^{-\alpha\xi}$. These are smooth, analytic functions of the single-electron coordinates, and—the point that matters here—they carry no φ dependence at all.

[MEASURED] The convergence study itself shows that the gap is not reachable along the directions the basis does span. Going from $N = 27$ to $N = 46$ to $N = 72$ moves the energy 92.2% → 92.4% → 92.4%: 0.34 mHa for 2.7× the functions. Whatever is missing, more σ functions do not supply it. Sections VI B and VII identify it and remove it.

B. The electron-electron cusp is real, but it is not this gap

The exact two-electron wavefunction $\Psi(\mathbf{r}_1, \mathbf{r}_2)$ satisfies the Kato cusp condition [16] at the electron-electron coalescence point $r_{12} = 0$:

$$\left.\frac{\partial \bar{\Psi}}{\partial r_{12}}\right|_{r_{12}=0} = \frac{1}{2}\,\bar{\Psi}(r_{12} = 0), \tag{29}$$

where $\bar{\Psi}$ is the spherically averaged wavefunction. This cusp produces a discontinuity in the first derivative of Ψ with respect to r_{12} and, more fundamentally, introduces non-analytic dependence on r_{12}.

The Fock expansion [17, 18] of the exact wavefunction near the two-electron coalescence point has the form

$$\Psi = \sum_{n=0}^{\infty} \sum_{k=0}^{\lfloor n/2 \rfloor} R^{n/2} \,(\ln R)^k \, f_{nk}(\hat{\Omega}), \tag{30}$$

where $R = (r_1^2 + r_2^2)^{1/2}$ is the hyperradius and $\hat{\Omega}$ represents the hyperangular coordinates. The $R^{1/2}$ and $R \ln R$ terms are fundamentally non-analytic: no finite sum of polynomial×exponential functions in prolate spheroidal coordinates can represent them *exactly*.

TABLE I. Convergence of H_2 ground-state energy at $R = 1.4011$ bohr. N is the number of basis functions. $D_e\%$ is the fraction of the exact dissociation energy $D_e = 0.174475$ Ha recovered. $\Delta E = E_{\text{Neumann}} - E_{\text{numerical}}$.

Config	N	E_{num} (Ha)	$D_e\%_{\text{num}}$	t_{num} (s)	E_{Neu} (Ha)	$D_e\%_{\text{Neu}}$	t_{Neu} (s)	ΔE (Ha)
$j=0$, $l=0$	1	−1.073192	41.9	1	−1.107907	61.8	3	−0.0347
$j=1$, $l=0$	3	−1.083438	47.8	8	−1.115282	66.1	9	−0.0318
$j=1$, $l=1$	6	−1.103485	59.3	32	−1.129606	74.3	32	−0.0261
$j=2$, $l=1$	12	−1.107960	61.9	152	−1.132373	75.9	131	−0.0244
$j=2$, $l=2$	27	−1.138836	79.6	767	−1.160961	92.2	705	−0.0221
$j=3$, $l=2$	46	−1.139329	79.8	2248	−1.161162	92.4	1907	−0.0218
$j=3$, $l=3$	72	−1.139792	80.1	6076	−1.161304	92.4	5271	−0.0215

TABLE II. Grid convergence of numerical V_{ee} for the 6-function basis at $R = 1.4011$ bohr, $\alpha = 1.0$.

Grid ($N_\xi \times N_\eta \times N_\phi$)	$D_e\%$	t (s)
$12 \times 8 \times 8$	66.0	1
$20 \times 14 \times 16$	76.2	6
$30 \times 20 \times 24$	81.4	45
$40 \times 26 \times 16$	83.7	83
$60 \times 40 \times 16$	85.7	447

[WITHDRAWN] Earlier versions of this paper concluded from this that the cusp is what caps the calculation at 92.4%. It is not, and the argument was never sound: non-analyticity implies slow convergence of a partial-wave expansion, not a floor at 7.6%. Two calculations settle it. Tao, McCurdy and Rescigno [7], working in these same prolate spheroidal coordinates with a polynomial angular basis and no r_{12} dependence anywhere in it, carry out an accurate two-electron H_2 calculation in these same coordinates. (Their odd-m radial functions do carry an explicit $(\xi^2 - 1)^{1/2}$, which they introduce for the non-analytic behaviour at $\xi = 1$; that is a one-electron boundary condition on the focal axis, the analogue of the $\sin\theta$ behaviour of ordinary spherical harmonics at odd m, and is unrelated to r_{12}. The point stands as stated: nothing in their basis represents the *interelectronic* non-analyticity.) And the calculation reported in Sec. VII below, run in the basis of Eq. (4) with the same Neumann kernel and likewise no r_{12} dependence, reaches 99.1%. Neither could exist if representing the cusp were the binding constraint at this accuracy. What the cusp does cost is the slow tail of the partial-wave series; at $|m| \le 1$ and $l \le 3$ that residue is under one percent of D_e.

C. Why r_{12} basis functions help only partially

The traditional approach to the cusp problem is to include explicit r_{12} dependence in the basis, as in the classic James–Coolidge [13] and Kolos–Wolniewicz [14] wavefunctions. We tested this by extending the basis with r_{12}^p ($p = 1, 2$) factors. The results are instructive: with 18 functions including r_{12}^2 and angular terms ($l \le 1$), the numerical V_{ee} reaches 86.8% D_e at the $20\times14\times16$ grid—but this *also* saturates with grid refinement. The r_{12} factors improve the wavefunction's ability to represent the cusp, but the V_{ee} integrals involving r_{12} are even harder to evaluate numerically (they become seven-dimensional integrals with an additional radial singularity).

The Neumann algebraic approach at 92.2% with $p = 0$ basis functions outperforms the r_{12} + numerical approach at 86.8% with $p \le 2$ basis functions. What that comparison establishes is narrower than it looks: *an exact V_{ee} beats r_{12} basis functions carrying grid error.* It says nothing about the cusp in either direction. (Earlier versions drew the stronger conclusion that "the cusp does not need r_{12}^p basis functions—it needs exact V_{ee}"; that goes with the rest of the withdrawn cusp reading, and Sec. VI B gives the cusp's actual cost here.) The r_{12} basis functions are a workaround for poor V_{ee} integration, not a solution to the underlying representation problem.

D. The missing channels

[MEASURED] The 7.6% gap is a *configuration-space* limitation, and an elementary one. A $^1\Sigma_g^+$ state requires the *total* azimuthal quantum number $M = m_1 + m_2$ to vanish. It does not require $m_1 = m_2 = 0$. The configurations π^2 ($m_1 = +1$, $m_2 = -1$), δ^2 ($m_1 = +2$, $m_2 = -2$), and so on all have $M = 0$ and are all $^1\Sigma_g^+$; they carry the *angular* part of the electron correlation, the left–right and in–out parts being what a σ-only space already describes. The basis Eq. (4) is φ-independent and therefore excludes every one of them, and Eq. (3) drops exactly the kernel terms that would couple them in. The two restrictions are consistent with each other, which is why the calculation converges cleanly to a wrong answer.

Section VII restores the channels and measures the result: 92.4% → 99.1% of D_e. What remains after that is ordinary basis incompleteness—higher l, higher n, and the δ channels. That remainder is ≈1.6 mHa in total, of which the δ channels account for ≈0.5 mHa on an independent Gaussian-basis estimate; the balance is not itemised here.

For completeness, the interelectron coordinate in pro-

late spheroidals is a function of all five relative variables:

$$r_{12}^2 = \frac{R^2}{4}\left[(\xi_1^2-1)(1-\eta_1^2)+(\xi_2^2-1)(1-\eta_2^2)-2\sqrt{(\xi_1^2-1)(1-\eta_1^2)(\xi_2^2-1)(1-\eta_2^2)}\cos(\varphi_1-\varphi_2)+(\xi_1\eta_1-\xi_2\eta_2)^2\right] \tag{31}$$

(The $\cos(\varphi_1-\varphi_2)$ factor was omitted in earlier versions of this equation, which therefore held only at $\varphi_1=\varphi_2$.)

Note the $\cos(\varphi_1-\varphi_2)$ dependence: the interelectron distance is not a function of $\xi_1,\eta_1,\xi_2,\eta_2$ alone. A φ-independent basis cannot resolve it, which is the same restriction stated in coordinates.

[**WITHDRAWN**] Earlier versions of this paper closed this section by identifying hyperspherical coordinates $(\rho,\theta,\hat{\Omega})$ as the natural geometry the gap called for, on the grounds that the coalescence cusp has a simple representation there. That inference rested on the cusp diagnosis withdrawn above and does not survive it. Hyperspherical coordinates remain the right setting for genuine three-body coalescence problems, and Paper 13 [5] uses them for helium on independent grounds; what is withdrawn is the claim that *this* calculation's residual motivates them.

VII. RESTORING THE AZIMUTHAL CHANNELS

[**MEASURED**] We extend the basis of Eq. (4) with an azimuthal quantum number,

$$u_{jl\mu}(\xi,\eta) = \xi^{\,j}\,\eta^{\,l}\,(\xi^2-1)^{\mu/2}(1-\eta^2)^{\mu/2}\,e^{-\alpha\xi}, \qquad \Phi_\mu = u\,u'\cos\mu(\varphi_1-\varphi_2), \tag{32}$$

pairing $m_1=+\mu$ with $m_2=-\mu$ through $\cos\mu(\varphi_1-\varphi_2)$—the $M=0$, ${}^1\Sigma_g^+$ combination, which for $\mu=1$ is the familiar $\pi_x\pi_x+\pi_y\pi_y$. At $\mu=0$ the factor u reduces to the one-electron factor of Eq. (4) and Φ_0 to its symmetrised product, so the $\mu=0$ sector is exactly the original basis. The overlap, kinetic and nuclear-attraction matrices remain *exact*—every integrand is a polynomial in ξ and η times $e^{-2\alpha\xi}$, evaluated against the moments A_n of Sec. III and elementary η moments—and all three are diagonal in μ. V_{ee} uses the full kernel Eq. (2); its $m\neq 0$ terms are what couple different μ.

The η integrals remain exact polynomial moments, because the kernel's $(1-\eta^2)^{m/2}$ and the basis's $(1-\eta^2)^{\mu/2}$ always combine to an integer power. The sum over l terminates exactly: integrating by parts m times (the boundary terms vanish since $(1-\eta^2)^s$ has a zero of order $s\geq m$ at $\eta=\pm1$) shows the moment is identically zero for $l>Q+2s-m$, with Q the combined η power and $s=(\mu_i+\mu_j+m)/2$. That rule must be imposed explicitly rather than left to cancellation: the Legendre derivative coefficients reach $\sim 10^{10}$, and their floating-point residue multiplied by the radial integral produced a -3×10^8 Ha artifact before the cutoff was enforced.

Table III gives the result. At the largest basis the azimuthal channels are worth +11.64 mHa, taking 92.42% → 99.1% of D_e. The $\mu=0$ column reproduces the Neumann column of Table I to 161, 1.6, 1.0, 6.7 and 58 μHa.

TABLE III. H_2 at $R=1.4011$ bohr with the azimuthal channels restored. Same basis family, same Neumann kernel, α scanned, canonical orthogonalization throughout — eigenvectors of S with eigenvalue below $10^{-11}\lambda_{\max}$ are discarded, and the surviving dimension is given in parentheses where it differs from N. $D_e^{\rm exact}=0.174475$ Ha.

$(j_{\max},l_{\max})$	N_σ	$D_e\%$ ($\mu=0$)	$N_{\lvert m\rvert\leq1}$	$D_e\%$ ($\lvert m\rvert\leq1$)
(1, 1)	6	74.19	12	81.63
(2, 1)	12	75.87	24	83.47
(2, 2)	27	92.25	54	98.96
(3, 2)	46	92.37	92 (76)	99.00
(3, 3)	72 (65)	92.42	144 (115)	99.09

Three controls make the reading unambiguous. *First*, the gain is not a basis-count effect: tripling the basis along the σ axis is worth 0.34 mHa (Sec. VI), while opening the azimuthal axis at fixed $(j_{\max},l_{\max})$ is worth 11.6 mHa. *Second*, an independent route agrees. A full CI in a Cartesian Gaussian basis—different functions, different integrals, different code—restricted to $m=0$ orbitals converges to 92.34%, within 0.2 mHa of this paper's σ-only value in a completely unrelated basis; releasing $|m|\leq1$ takes it to 99.10%, against 99.1% here—agreement well inside the stability envelope quoted below, and not to be read as a two-decimal coincidence. *Third*, the general-m kernel of Eq. (2) was checked pointwise against $1/|\mathbf{r}_1-\mathbf{r}_2|$, agreeing to better than 10^{-6} at well-separated ξ by $l=10$. That figure is a point evaluation, not a uniform bound: the Neumann series converges slowly when the two ξ are close, and the second-kind recursion used here degrades at larger l, so nearby geometries and higher truncations are worse. What the check establishes is the *form* of the prefactor—the superseded one errs by nine orders of magnitude at the same point—not an accuracy claim for the kernel.

Backing: `tests/test_paper12_azimuthal_channels.py`, whose guards are fire-tested against the specific wrong answers they exclude (a killed azimuthal coupling, the superseded kernel prefactor, a disabled canonical orthogonalization, and a truncation cap that would hide the selection rule).

[**MEASURED**] Both limits stated in earlier versions of this section have since been removed. The $\mu>0$ radial integrals are now available from the A_l, B_l, X_l recurrences of Sec. III generalised to the associated Legendre functions (module `geovac/neumann_vee_general_m.py`). The first-kind moment $A_l^{m,s}(p) = \int_1^\infty \xi^p(\xi^2-1)^s e^{-c\xi}\,[\,\mathrm{d}^m P_l/\mathrm{d}\xi^m\,]\,\mathrm{d}\xi$ is a closed-form polynomial contraction against the A_n; the second-kind moment $B_l^{m,s}$—whose integrand is singular as $(\xi-1)^{-m}$ at $\xi=1$, regularised by the weight $(\xi^2-1)^s$ kept *intact* rather than expanded into ξ-powers (the bare $\xi^p\,\mathrm{d}^m Q_l$ moment diverges for $m\geq1$)—is built by the associated-Legendre forward l-recurrence $(l-m+1)B_{l+1}(p) =$

$(2l+1)B_l(p+1)-(l+m)B_{l-1}(p)$, seeded at $l=m, m+1$; the ordered-ξ integral $X_l^{m,s}$ then follows from the same integration-by-parts split as the σ case. The engine reduces to `geovac.neumann_vee` at $m=0$ to $\sim 10^{-9}$ elementwise, matches a high-precision reference for $X_l^{m,s}$ to better than 10^{-9} through $l=8$, $m\le 4$, and reproduces the grid result at $(2,2)$, $|m|\le 1$ to the last printed digit. Its decisive property is *stability* in the δ ($|m|=2$) sector: the fourth-derivative Q_l cancellation near $\xi=1$ drives the differentiation route to a non-variational ≈ -14.5 Ha, whereas the recurrence build is variational and supplies the δ channels from the native prolate route—a $+0.25$ mHa gain at $(2,2)$, consistent with the ≈ 0.5 mHa Gaussian estimate (the $(2,2)$ δ subspace being the smaller). The engine is *fully quadrature-free*: the low-l $B_l^{m,s}$ seeds are closed-form by the same construction as the σ sector (Sec. III D). The obstruction that had kept these seeds numerical—the individual pole terms $d^kQ_0 \sim (\xi-1)^{-k}$ of d^mQ_l have divergent moments—dissolves because $s\ge m$ always holds ($s=(\mu_i+\mu_j+m)/2$ with $m\in\{\mu_i+\mu_j, |\mu_i-\mu_j|\}$): the intact weight $(\xi^2-1)^s=(\xi-1)^s(\xi+1)^s$ then *fully polynomialises* every pole ($k=m-a\le m\le s$), leaving a single log-moment against Q_0 whose primitive is Eqs. (21)–(23). The moment tables are still built in extended precision (the associated Legendre coefficients cancel over several orders), but no numerical integration remains. Backing: `tests/test_paper12_general_m_neumann.py`.

[**MEASURED**] Finally, a caution about the σ column itself. The overlap matrix of this basis family is strongly linearly dependent at high powers and a single α: measured $\mathrm{cond}(S)=2.6\times10^{14}$ at $(j,l)=(3,3)$, $\mu=0$ (the $N=72$ basis of Table I), rising to 2.0×10^{16} once $\mu=1$ doubles the basis. Beyond $\sim 10^{16}$ a direct generalised eigensolve is not trustworthy—at $N=144$ it returned -79 Ha—so Table III uses canonical orthogonalization throughout.

[**MEASURED**] Canonical orthogonalization *bounds* that pathology; it does not remove it, and we state the envelope rather than leave the reader to assume a converged four-figure number. At the $|m|\le 1$, $N=144$ basis the solver still returns non-variational values at a *majority* of α. Measured at the stated threshold over $\alpha\in[0.90,1.30]$, six of nine grid points fail. The failures are not marginal: $\alpha=1.15$ and 1.20 return -4.1 and -8.1 Ha, $\alpha=0.95$ returns -277 Ha, and $\alpha=1.00$—the natural default, and the value a reader is most likely to try first—returns -64.3 Ha. The three that survive give 99.03, 99.09 and 99.05% at $\alpha=1.10$, 1.25, 1.30. So the reported value is the best *variational* point of an α scan, and points failing the variational test are discarded—a selection the reader should see, not infer. The value also moves with the discard threshold: at $\alpha=1.25$ it reads 99.14% at 10^{-12}, 99.09% at 10^{-11}, 98.99% at 10^{-10} and 98.41% at 10^{-8}, and is non-variational below $\sim 10^{-12}$. The honest statement is therefore 99.0–99.1%, with 99.09% the best variational value at the stated parameters.

[**SCOPE 2026-09-14, corrected twice**] Two earlier versions of this passage overstated how stable that number is. The first called the conclusion "insensitive to the choice" [retracted 2026-09-14: p12-envelope-insensitive] and then quoted the figure that refutes it. The second replaced it with a range, "95.5–99.1% across the grid", and called the qualitative conclusion "robust across the grid" [retracted 2026-09-14: p12-grid-floor-955]. Both are wrong, and the second is wrong in the direction that flatters the result.

95.5% is the cell at $\alpha=1.10$, threshold 10^{-8}. It is the weakest variational value only if α is restricted to ≥ 1.10—which excludes precisely the low-α half this paragraph has just identified as pathological. Over the full declared range $\alpha\in[0.90,1.30]$ the weakest variational value is 76.11%, at $\alpha=0.90$ and threshold 10^{-8}; and at that same cell the σ-only solve returns 91.06%, so opening the azimuthal channels there makes the answer *worse* by fifteen points. The qualitative conclusion is therefore *not* robust pointwise across the grid, and this paper does not claim that it is.

What the result rests on instead, stated exactly: at the tight discard thresholds, where the generalized eigenproblem is actually being solved rather than regularized, every surviving variational point lies between 99.03 and 99.14%; and the independent Gaussian-basis route, which is well conditioned and needs no discard threshold at all, gives 99.10%. The claim is the agreement of those two. It is not a property of the whole $(\alpha,\text{threshold})$ grid, most of which is conditioning noise. The -1.161304 Ha quoted in Table I sits 58 μHa below the conditioned value, i.e. the last two of its six decimals reflect linear dependence rather than physics. No conclusion in this paper turns on them. That the number is conditioning-*limited* rather than physics-limited is shown directly in Sec. VII A: re-basing this same span removes the linear dependence and the energy climbs to chemical accuracy.

A. The monomial cap is conditioning, not a ceiling

[**MEASURED**] The 99.1% above is the ceiling of the *monomial* radial factor ξ^j, not of the prolate geometry or the electron-electron cusp. The set $\{\xi^j e^{-\alpha\xi}\}$ is a Hankel moment problem, exponentially ill-conditioned in the degree: the overlap reaches $\mathrm{cond}=1.8\times10^{16}$ already at $(3,3)$, $|m|\le1$ and 4.1×10^{26} at $(5,5)$, so the basis cannot be pushed further in double precision. But the monomials $\{\xi^j\}_{j\le J}$ and, e.g., the Laguerre polynomials $\{L_j(2\alpha(\xi-1))\}_{j\le J}$ span the *same* degree-J space, so re-basing onto an orthogonal-polynomial family is an *exact* change of basis: it leaves the span—and hence the exact-arithmetic energy—unchanged, and only the conditioning changes.

[**MEASURED**] Because the congruence $CSC^{\top}$ amplifies the 10^{-16} entry error by $\|C\|^2$ and the high-degree monomial matrices carry $\mathrm{cond}\sim10^{26}$, a float64 change of basis is corrupted—it breaks for *both* families at every truncation tested. Every matrix entry that passes

through the monomial basis is therefore built in extended precision, the well-conditioned orthogonal matrices are downcast to float64, and the generalized eigenproblem is solved on the unit-normalized (correlation) matrices—a diagonal congruence that leaves the eigenvalues unchanged but removes the norm-spread inflation of cond.

[**MEASURED 2026-09-18**] *Two condition numbers appear in this section and they answer different questions; conflating them is a live trap.* The raw cond figures above (1.8×10^{16}, 4.1×10^{26}) are the right metric for the $\|C\|^2$ congruence-corruption argument, which is about entry error amplified by the change of basis. They are *not* the metric the eigensolve sees: that is the unit-normalized cond, and the two can differ by six orders on the same matrix. Measured at $(3,3)$, $|m| \le 1$: raw 3.0×10^{15} against normalized 1.1×10^{11}. The practical consequence is not academic—a raw comparison was used to conclude that a two-exponent radial set carried a 47× conditioning penalty, whereas on the governing metric the same basis is 544× *better* conditioned than the single-exponent one, because two distinct decay rates make the functions less linearly dependent rather than more. Quote raw cond when arguing about a congruence and normalized cond when arguing about a solve. With that pipeline the energy climbs monotonically and variationally past the monomial wall (Table IV): opening the δ channels on the re-based basis reaches 99.71% of D_e at $(4,4)+\delta$ and 99.77% (0.41 mHa) at $(5,5)+\delta$—inside chemical accuracy, a four-fold reduction of the residual from the monomial 99.09%, every point containing its predecessor and lying below it.

[**MEASURED**] That ladder is computed at a *fixed* $\alpha = 1.0$, which is what makes it a ladder: the statement that each point contains its predecessor and lies below it holds only at a consistent exponent, so Table IV keeps α fixed throughout. But α is a variational parameter, and 1.0 is not its optimum. Scanning it at the largest truncation gives 0.406, 0.346, 0.32 and 0.334 mHa at $\alpha =$ 1.00, 1.20, 1.40 and 1.50, so the minimum is bracketed at $\alpha = 1.40$ and the best variational result this basis supports is 99.81% of D_e (0.32 mHa)—better than the $\alpha = 1.0$ endpoint by 0.082 mHa, a fifth of that residual. The improvement is not a trade: the optimal point is variational, retains all 1944 functions, and is *more* tightly conditioned than $\alpha = 1.0$ (4.9×10^{10} against 9.4×10^{10}). Both numbers are therefore reported, and they are not competing: 99.77% is the endpoint of the fixed-α ladder, 99.81% is the basis's best variational point.

[**MEASURED 2026-09-18**] *What the remaining residual is not.* Three knobs of this basis have now been measured against the $|m| \le 1$ ladder, one axis at a time, and none of them reaches it: radial degree +1 buys 34 μHa, angular degree +1 buys 8 μHa, and re-optimizing the single exponent buys 0.056 mHa—together about 9% of that rung's 1.096 mHa residual. A *two-block* exponent set (low degrees at a_1, high degrees at a_2, at equal function count) does not reach it either: +0.005 mHa at $(4,4)$ and −0.009 mHa at $(3,3)$, and the $(4,4)$ optimum is a *plateau* rather than a point—1.107, 1.108, 1.110 mHa across three quite different (a_1, a_2) pairs, flat to 3 μHa—which is direct evidence that a second exponent has no work to do at this basis size. That last result also disposes of the natural reading of the drift in α_{opt} with basis size ($\sim$1.20 at $(4,4)$, 1.40 at $(5,5)$): the drift is real but it is not relievable strain. We state this as the *elimination* of three knobs, not as a measurement of the cusp: what is established is that more polynomial degree and more exponents do not close the gap in this family, which is weaker than—and should not be quoted as—a demonstration that the remainder *is* the electron-electron cusp.

TABLE IV. H_2 at $R = 1.4011$ bohr, re-based to orthogonal polynomials (Laguerre×Legendre and the μ-adapted Gegenbauer give the identical energy). Same span as the monomial basis of Table III, re-conditioned; $\alpha = 1.0$, the extended-precision V_{ee} build and unit-normalized solve of the text. $D_e^{\text{exact}} = 0.174475$ Ha.

basis / channels	N	D_e%
monomial ceiling ($\|m\| \le 1$)	144	99.09
re-based $(3,3)$, $\|m\| \le 1$	256	99.22
re-based $(4,4)+\delta$	975	99.71
re-based $(5,5)+\delta$	1944	99.77

[**MEASURED**] The extended-precision requirement above constrains the *change of basis*, not the orthogonal matrices themselves, and so does not apply to a matrix that is never built in the monomial basis at all. The one-body operators admit such a construction: every one-electron block factorises as radial × angular, so S and H_1 can be assembled directly in the orthogonal basis from one-dimensional blocks small enough to evaluate exactly, after which the $O(N^2)$ two-electron assembly is ordinary double precision. Built that way they agree with the extended-precision route re-based to 6×10^{-16} of the matrix scale at $(5,5)+\delta$ in both families, and every energy in Table IV is unchanged. Extended precision is therefore not a requirement of the one-body half at all: its one-dimensional blocks are evaluated exactly because they are small enough for that to cost nothing, not because a congruence forces it. What does still force it is the V_{ee} re-basing, which accordingly accounts for essentially the whole cost of the calculation.

[**MEASURED**] What the residual $\sim$0.4 mHa is *not* is more azimuthal channels: φ buys under +0.06 pp on top of δ. What it *is* remains unattributed, and per the elimination statement above it must not be quoted as the electron–electron cusp: radial completeness and the cusp are the two standing candidates, both converging as the slow L^{-3} partial-wave crawl, and the knob elimination does not separate them. A literal 99.9% therefore needs a much larger basis or explicit correlation (geminals); what is established is that a *radial* re-basing within this family does not reach it.

[**MEASURED**] The μ-weight-adapted family—the generalized Laguerre $L_n^{(\mu)}$ for the $(\xi^2-1)^{\mu/2}$ weight and

the Gegenbauer $C_n^{(\mu+1/2)}$ for $(1-\eta^2)^{\mu/2}$, reducing to plain Laguerre×Legendre at $\mu = 0$—spans the identical space (the energy agrees to < 1 μHa at every truncation; at $(5,5)+\delta$ and the optimal $\alpha = 1.40$ the two families agree to all seven printed digits, $E = -1.1741513$ Ha) at a normalized condition number *six* orders of magnitude smaller (3.7×10^4 vs 4.9×10^{10} at $(5,5)+\delta$, $\alpha = 1.40$; 9.1×10^4 vs 9.4×10^{10} at $\alpha = 1.0$), which is what keeps the downcast solve trustworthy at large truncation. **[MEASURED]** Earlier versions of this sentence read "two to three orders of magnitude" while quoting the $(5,5)+\delta$ pair, whose ratio is 1.0×10^6; the "two to three" is the 326× of the $(3,3)$, $|m|\le 1$ case and does not describe the figures it was attached to. It does *not* reach a given accuracy with fewer functions: at equal degree the span, and hence the energy, is the same—the gain is conditioning alone. A per-*function* exponent instead (making the set L^2-orthonormal) would break completeness and plateau the accuracy, and is not used.

Backing: module `geovac/prolate_recondition.py`, tests `tests/test_paper12_recondition.py`.

B. Explicit correlation, evaluated algebraically

The knob-elimination above closes with an explicit alternative: a literal 99.9% "needs a much larger basis or explicit correlation (geminals)." The second route is available inside this paper's own machinery, and it does not leave the algebraic (quadrature-free) discipline of Sec. II. Multiply each product function by a power of the interelectronic distance, $\Phi = g(\xi_1,\eta_1;\xi_2,\eta_2)\, r_{12}^p$ with $p \in \{0,1\}$ (a James–Coolidge basis in prolate coordinates), and every resulting matrix element—overlap, V_{ne}, V_{ee}, and kinetic—reduces to the *same* algebraic objects already built for V_{ee}.

[MEASURED 2026-09-20] The reduction is exact. Writing $r_{12}^2 = (R/2)^2(A - B\cos\Delta\varphi)$ with A, B the standard prolate combinations, every *even* power of r_{12} is a polynomial in the separated coordinates (its azimuthal average is a finite sum of the one-electron moments of Sec. II), and every *odd* power is $r_{12}^{2s+1} = (R/2)^{2s+2}(A - B\cos\Delta\varphi)^{s+1}/r_{12}$, whose azimuthal average is the same $A\,K_0 - B\,K_1$ combination of Neumann kernels—the $m = 0$ and $m = 1$ channels of the general-m V_{ee} table of Sec. II—that the Coulomb term itself uses. The one element that looks harder, the kinetic coupling between a $p = 0$ and a $p = 1$ function, collapses: integrating $\frac{1}{2}\langle\nabla g_u\cdot\nabla(g_v r_{12})\rangle$ by parts cancels the gradient-of-r_{12} term against half of itself and leaves

$$T_{p_0\times p_1} \;=\; -\tfrac{1}{2}\int r_{12}\, g_v\, (\nabla^2 g_u)\, dV, \tag{33}$$

a pure odd-r_{12}^1 element (no ∇r_{12}, no non-analytic term) with $\nabla^2 g_u$ a polynomial numerator, routed through the same kernels. Each block was validated against direct quadrature to $\sim 10^{-4}$ (grid-limited).

[MEASURED 2026-09-20] On the σ ($\mu = 0$) gerade basis this reaches 99.97% of D_e (0.053 mHa) at $(j_{\max}, l_{\max}) = (3,4)$, $n = 416$, $\alpha = 1.0$—a sixfold reduction of the residual from the re-based CI's 0.32 mHa, obtained with exact algebraic integrals (no quadrature at any stage). The energy descends monotonically from above along nested truncations (-20.7, -0.167, -0.079 mHa at $n = 18$, 90, 160), and the extended precision of Sec. VII A is again load-bearing: by $n \simeq 400$ the overlap conditioning reaches $\sim 10^{19}$ and the double-precision solve discards genuine directions (16 μHa above the extended-precision energy), while the mpf-orthogonalized solve retains them.

[MEASURED 2026-09-20] This constructively settles what the Sec. VII A residual *is*. It is electron–electron correlation, captured by ordinary integer powers of r_{12}—not the non-analytic $r_{12}^{1/2}$, $r_{12}\ln r_{12}$ terms this paper once invoked (Abstract, **[WITHDRAWN]**). The explicit-r_{12} energy then *plateaus* near 0.05 mHa: at $p \le 1$ and a single α the overlap becomes linearly dependent (cond $\sim 10^{19}$ with essentially all directions retained), i.e. that space is nearly spanned; a variational α scan gains only $\sim 2\,\mu$Ha (its optimum is $\alpha \approx 1.0$–1.2, and the CI optimum $\alpha = 1.40$ is *worse* here, because explicit r_{12} already supplies the correlation a contracted exponent was compensating for). The microhartree regime is therefore not an α or radial-degree question but a higher-r_{12}-power ($p \ge 2$) one, which the same $A\,K_0 - B\,K_1$ reduction extends to (with additional $m \ge 2$ channels). We report the $p \le 1$ result and its measured ceiling, not a microhartree claim.

[MEASURED] The construction is not specific to H_2. Because the r_{12} kernels touch only the electron coordinates, the whole reduction is charge-independent: the sole term that changes for a *heteronuclear* two-electron diatomic is V_{ne}, whose per-electron form $(2/R)[(Z_A + Z_B)\xi + (Z_B - Z_A)\eta]/(\xi^2 - \eta^2)$ stays polynomial after the Jacobian (the $(Z_B - Z_A)\,\eta$ term breaks the $\eta \to -\eta$ symmetry, so both angular parities enter and V_{ne} couples them). Applied to HeH^+ at the reference geometry $R_e = 1.463\,a_0$ the same engine converges variationally to 0.9 mHa of the Kolos–Peek reference [15] (well depth 16456 cm^{-1}, $E_{\rm BO}(R_e) = -2.9787$ Ha) at a modest basis, and—consistently with the H_2 plateau above—the lever there is the *angular* basis, not the exponent (a two-block exponent gains ~ 0.3 mHa, one more angular shell ~ 11 mHa). Scanning R locates the bond length as well: $R_{\rm eq}$ converges from -4.5% at $l_{\max} = 2$ to -0.5% at $l_{\max} = 3$ (1.456 vs $1.463\,a_0$), by the *same* angular lever—so the engine reproduces the heteronuclear *geometry*, not only the energy, and the residual is heteronuclear polarization (an angular effect). This is the two-center, two-different-charge analogue of the H_2 convergence, and the precedent for the frozen-core LiH extension.

[MEASURED 2026-09-21] The reduction is also not specific to two electrons, and this places the natural boundary of the quadrature-free, *resolution-of-identity–free* discipline. For a James–Coolidge ansatz $\Psi = F\,\Phi$

with $F = \sum_{p<q} f_{pq}$ multiplicative, the variational kernel $\langle\Phi|FHF|\Phi\rangle$ stays at most three-body through $N = 3$: every term of the operator product closes with closed-form angular rules (a shared-vertex rule fixing scalar legs to $L = 0$, and the triangle rule $\langle P_a(12)P_b(13)P_c(23)\rangle = \delta_{abc}/(2a+1)^2$), so the three- and four-electron integrals that production Gaussian R_{12}/F_{12} theory treats with a resolution of identity are here evaluated *exactly*, with no such approximation. On one centre this reduction is classical: integrating out one electron, or expanding the r_{ij} factors across a shared electron with an addition theorem, is how atomic Hylleraas-CI evaluates its three- and four-electron integrals, exactly over Slater bases for four electrons [21–23]; the triangle rule is the standard one-centre angular collapse; and RI-free routes exist for Gaussian geminals [24] and by numerical quadrature [25]. Two-center explicitly correlated integrals over exponential geminals are known for two electrons [26, 27]. What we have not found in the literature (a search of the Hylleraas-CI, F_{12} and two-center correlated-integral work, some of which we could not reach) is the two-center extension reported below: exponential geminals between prolate-spheroidal densities, with the Coulomb by the general-m Neumann expansion, inside a four-electron diatomic energy on a multi-determinant reference, without a resolution of identity. At $N = 4$ a genuinely four-body term first appears—the disjoint-pair Coulomb chain $f_{12}\, f_{34}\, r_{13}^{-1}$, with the two correlation edges bridged by one Coulomb, four distinct electrons (only the two-body Coulomb can bridge disjoint pairs; the kinetic gradients are pair-local and V_{ne} is one-body). This term is nonetheless still *exact*, still RI-free, and still *reducible*: the bridging-Coulomb multipole sum terminates, bounded by the finite orbital angular content and not by the Coulomb: at $L \le 2\, l_{\mathrm{bridge}}$ for s-type leaves (measured $s \to 0$, $p \to 2$, $d \to 4$), and in general at $L \le 2\,(l_{\mathrm{bridge}} + l_{\mathrm{leaf}})$—for p-type leaves the $L = 4$ channel that the s-leaf bound would discard is measurably nonzero (a 24σ Monte-Carlo detection, 2026-09-22), so the general bound must be used whenever the leaves are not isotropic; the sum remains finite, so the RI-free property is untouched—and the Legendre addition theorem factorizes the chain across the bridge into a finite (L, M) contraction of vertex kernels—the chain analogue of the triangle contraction. A native evaluation (deterministic quadrature on the reduced form, not a closed formula) was validated against brute-force 12-dimensional Monte-Carlo integration to 2×10^{-4} on the Be-relevant four-electron integral ($2p$ bridge, $1s$ leaves), and, with the antisymmetric determinant (exchange chains), at $L = 0$ with all-s leaves (relative 9×10^{-5} on a 2026-09-22 re-run; higher-L exchange chains have not been run). As a proof of concept it was exercised inside a minimal-basis Be R_{12}-CI, where a single short-range geminal captures $\sim 19\%$ of the correlation energy variationally with the four-body term evaluated by this reduction. The residual hard walls are on separate axes—an explicit four-body correlation operator is a four-body Pauli string in a qubit encoding, and the non-Hermitian transcorrelated route carries vector legs that do not reduce this way—but the classical exactness axis, on which production Gaussian F_{12} uses a resolution of identity, extends past $N = 3$ here, as it does in atomic Hylleraas-CI [21]. This is a diagnostic result at proof-of-concept resolution, not a spectroscopic $N = 4$ claim.

[**MEASURED 2026-09-21**] Two extensions sharpen this boundary and carry the RI-free discipline from a standalone integral into a genuine many-electron correlation calculation. *First*, the four-body bridge is not tied to the single-center atom: evaluated between two-center prolate densities it reduces by the *same* construction—for isotropic ($1s$-type) leaves a one-dimensional radial leaf dressing, in general the two-center Neumann potential of the leaf density, followed by a two-center prolate Neumann Coulomb between the dressed densities—with the self-Coulomb step anchored to the closed form $5\alpha/8$. The σ channel reproduces an independent twelve-dimensional Monte-Carlo evaluation to 1.5×10^{-4} (the Monte-Carlo scatter, not the reduction, sets this figure); the π and δ channels were validated to $\sim 2 \times 10^{-5}$ on an azimuthally modulated model density $\rho_{1s}(1 + b_1 \cos\phi + b_2 \cos 2\phi)$, which exercises the general-m Neumann bridge but is not an $m \neq 0$ orbital density; the assembled energies below are σ-only. *Second*, the same machinery evaluates the many-body matrix elements of an explicit-r_{12} CI on a two-center four-electron reference ($\Phi_0 = |1s_A^2\, 1s_B^2|$, a proof-of-concept ionic reference, not a spectroscopic LiH) with no resolution of identity, each element checked against an independent variational Monte-Carlo ground truth. The entire off-diagonal coupling $h = h_T + h_{V_{ne}} + h_{V_{ee}}$ reduces to grid integrals of dressed one-electron densities and matches the Monte-Carlo values to within their statistical error: the kinetic part via an integration-by-parts identity that collapses the gradient coupling $\nabla F \cdot \nabla \Phi_0$ to a *scalar* screened-Coulomb (Yukawa) expectation ($\nabla^2 f = \gamma^2 f - 2\gamma\, e^{-\gamma r}/r$), the Coulomb part via the two-center Neumann *potential* of the transition density; the diagonal one-body moment $\langle \chi^2 V_{ne}\rangle$ ($\chi = F - \bar{F}$) reduces by a block decomposition $F = S + I$ (intra/inter) of the reference density with all three ($\langle S^2\rangle$, $\langle SI\rangle$, $\langle I^2\rangle$) parts Monte-Carlo–validated. Crucially, the four-body bridge appears *inside* the diagonal V_{ee} moment as $\langle f_{12} f_{34}\, r_{13}^{-1}\rangle = \frac{1}{4}\, I_{\mathrm{coul}}[S^f, S^f]$—the Coulomb self-interaction of the f-dressed one-electron slice, via the same Neumann potential—and again matches Monte-Carlo to within statistical error, so the RI-free treatment of the genuinely four-body content is demonstrated not merely as an isolated integral but inside a real correlation quantity. (A technical enabler: the screened-Coulomb dressing fields needed here are obtained without a Yukawa-specific Green's function, as $\Psi^Y = \Psi^{\mathrm{coul}} - \Psi^{\mathrm{s}}$ with the non-singular kernel $s(d) = (1 - e^{-\gamma d})/d$.)

[**MEASURED 2026-09-21**] The diagonal $\langle \chi^2 V_{ee}\rangle$ closes the analytic energy. Its one genuinely non-separable class is a two-center *three-body triangle* $\langle f_{ik} f_{jk}\, r_{ij}^{-1}\rangle$ (two correlation edges sharing a vertex k,

closed by a Coulomb on the opposite edge), whose down-integration yields a non-separable two-point kernel $H_2(1,2) = \int \mu(3) f(r_{13}) f(r_{23})\, d^3 r_3$. It is reduced RI-free by resolving H_2 into azimuthal Fourier components $H_2^{(m)}$, low-rank expanding each ($H_2^{(m)} = \sum_r \lambda_r\, v_r \otimes v_r$), and contracting mode-by-mode with the general-m prolate Neumann Coulomb—the same π/δ machinery built above. Both block placements (the Coulomb on the intra edge, so the shared vertex carries the marginal density; or on an inter edge, so the shared vertex carries an AO-pair component) reduce identically and reproduce Monte-Carlo to 0.2–0.4σ. Summed over all 216 interaction triples with their correct multiplicities, the analytic $\langle F^2 V_{ee}\rangle$ matches Monte-Carlo to $\sim 10^{-3}$ across all six intra/inter product groups, giving $g_{V_{ee}} = \langle \chi^2 V_{ee}\rangle = +0.538$ (+0.536 Monte-Carlo). The kinetic diagonal needs no vector ∇f dressing: the same integration-by-parts identity collapses $g_{T2} + g_{T3}$ to $-\gamma^2\sigma^2 + 2\gamma\,\mathrm{Cov}[F, \sum_{i<j} e^{-\gamma r_{ij}}/r_{ij}]$, leaving only a drift^2 moment (a grid-consistent block reduction over the gradient density), so $g_T = +1.394$ (+1.386 Monte-Carlo).

Assembling the fully analytic, RI-free 2×2 then gives, for the geminal $f = e^{-\gamma r}$, $E_{R12} = -7.942$ Ha—below the reference $E_0 = -7.888$ and above the exact -8.070 (variational), and matching the same-geminal Monte-Carlo 2×2 to 0.8 mHa. The framework is geminal-agnostic—only the kernels change—and the *cusp-correct* geminal $f = r\, e^{-\gamma r}$ (whose $\nabla^2 f = \gamma^2 f - 4\gamma e^{-\gamma r} + 2e^{-\gamma r}/r$ adds one term to the IBP-collapsed g_T) reproduces the headline proof-of-concept energy analytically: every reduced element matches its Monte-Carlo target below 1% (e.g. $\bar{F} = 3.570$, $\sigma^2 = 0.128$, $h = -0.103$), giving an assembled correlation lowering of -29 mHa (Monte-Carlo -29.5 mHa) and $E_{R12} = -7.917$ Ha. The one element that initially missed—the Yukawa covariance $\mathrm{Cov}[F, \sum e^{-\gamma r}/r]$, a small difference amplified by the long-range-geminal cancellation $\bar{F} = 3.570$—is closed by evaluating the isotropic screened-Coulomb dressings from the exact closed-radial Yukawa potential rather than the Neumann-minus-smooth field, exactly the distinction the Coulomb sector already draws between its closed Hartree potential and the two-center Neumann potential; this carries the covariance from -0.079 to -0.103 (Monte-Carlo -0.102), so the cusp-correct energy is reached fully analytically and RI-free with no residual quadrature caveat. The genuinely four-body and three-body content thus lives, RI-free, *inside* a two-center four-electron correlation energy—terms that production F_{12} methods treat with a resolution of identity. This remains a proof-of-concept (ionic single-ζ reference, not a spectroscopic LiH): the demonstration is the RI-free reduction, not the correlation recovered.

[AUDITED 2026-09-22] Two clarifications fix the standing of these reductions. *Tier.* The bridge factorization is a *validated reduction*, not a theorem: an exact re-expression of one operator term through three standard identities (the addition theorem or the Neumann expansion across the bridge, and the radial convolution of each leaf), stated here and checked numerically at the levels quoted, but never written out as a derivation, and carrying no stated hypotheses under the name "bridge addition theorem" that earlier sprint records attached to it. Its extension beyond four electrons is conjectural. Note that at the linear-in-F ansatz every term of $\langle\Phi|FHF|\Phi\rangle$ carries at most three correlation edges, so no term is ever more than four-body-connected for *any* N (the known Hylleraas-CI count for one r_{ij} per term [23]); what a larger system or a many-determinant reference changes is the *density*—leaves that are no longer isotropic, a reference that no longer factorizes into closed-shell blocks—and at the time of the audit that general path had been checked only as a Monte-Carlo–versus–Monte-Carlo factorization test at the 5% level (it has since been checked directly; see the build note below). Every deterministic validation of the audit used the block-factorized closed-shell reference $|\Phi_0|^2 = D(1,2)\, D(3,4)$, which was the enabling assumption of its spectator-integration bookkeeping. *Reference generality.* Read at the level of the code rather than the proof-of-concept, the reductions are not tied to the two-orbital reference: the bridge, the triangle kernel, the two-center Neumann potential, and the integration-by-parts identities each act on a single separable product of one-electron pair densities supplied as an argument, and the density of any four-electron CI is a finite sum of such products, so the reference enters only through a coefficient tensor over pair indices. The many-determinant extension is therefore unbuilt bookkeeping and a K^4 contraction cost, not an obstruction—and running it on a multi-orbital reference is precisely the test of the general, non-isotropic-leaf path, which the build note below now reports.

[BUILD IN PROGRESS 2026-09-22] The coefficient-tensor bookkeeping of the preceding paragraph has since been constructed for a twelve-orbital, σ-only CI reference (the core-enriched prolate CI of Paper 19, bond $(J, L) = (2, 1)$ plus three Li core exponents, $E = -8.02335$ Ha), rotated to its natural orbitals and truncated at $|c| > 10^{-3}$ to 74 determinants (+0.44 mHa above the full CI). The pair-index tensor over the 78 unordered orbital pairs has 38,786 nonzero entries, and its spectator traces reproduce the one- and two-particle reduced density matrices of the truncated vector to 2×10^{-16}, so the spectator-integration step that the closed-shell validations took from $|\Phi_0|^2 = D(1,2)\, D(3,4)$ is now exact for a general CI. The same construction exposed a quadrature defect that the closed-shell reference had masked: the proof-of-concept engine's prolate–Neumann Coulomb potential integrates the ordered kernel $P_l(\xi_<) Q_l(\xi_>)$ across its diagonal kink with cumulative Gauss–Legendre sums, an $O(N_\xi^{-2})$ error that is $+5 \times 10^{-3}$ relative on the Li $1s$ self-Coulomb and +8 mHa on $\langle V_{ee}\rangle$ of the CI reference at the engine's 72 ξ-nodes (the kinetic and nuclear-attraction grid integrals agree with the integral engine to 10^{-7} Ha, and the $\zeta = 4.5$ core density is resolved to 10^{-11}, so the defect is the kernel's quadrature, not the grid's resolution of the density). The closed-shell

engine is only mildly exposed because its isotropic pieces use a closed-form Hartree potential; the general-density path is fully exposed. The remedy is an exact ordered-integral operator on the same grid: the η-moments of every σ density are entire in ξ and are carried by the barycentric interpolant through the 72 Gauss–Legendre nodes, so the two partial integrals against $P_l^m(\xi')$ and $Q_l^m(\xi')$ can be taken exactly relative to that interpolant (one Gauss rule for the polynomial P side, geometrically graded panels toward the logarithmic endpoint on the Q side), and the result is a precomputed 72×72 operator per (l, m). It reproduces the $5\zeta/8$ self-Coulomb of an isotropic $1s$ density to 2×10^{-13} at $\zeta = 1.0$, 1.6, 2.6875 and 4.5 (the legacy sum is $+1.7 \times 10^{-3}$ to $+8.5 \times 10^{-3}$ off), the two-center Hartree closed forms to 7×10^{-14}, and solid-harmonic closed forms for $m = 0$–4 to 3×10^{-13}; with it, the grid kinetic, nuclear-attraction and electron-repulsion energies of the 74-determinant reference agree with the integral engine to 10^{-6} mHa, and the legacy path is bit-identical to the committed engine (fast guard `tests/test_lih_r12ci_neumann_exact.py`). On the corrected kernel the closed-shell proof-of-concept energies move by -0.4 mHa (cusp-correct geminal, to -7.9172 Ha, whose correlation energy -29.4 mHa now sits closer to the Monte-Carlo -29.5) and by -1.6 mHa (exponential geminal, to -7.9436 Ha, now 2.4 mHa below the same-geminal Monte-Carlo 2×2 value of -7.9412): the 0.8 mHa agreement quoted above for the exponential geminal was in part a cancellation between this Coulomb-kernel bias and the remaining coarse-grid errors of the f-kernel dressings, which the engine's own source recorded at the time and which the many-determinant build quantifies next. The banked closed-shell numbers are left on the legacy path, where the regression tests pin them.

[**VALIDATED 2026-09-23**] The generality test of the audit paragraph has now been run. On the exact operator, the four-body bridge integral $\langle \rho_1 \rho_2 \rho_3 \rho_4 \, f_{12} f_{34} / r_{13} \rangle$ was evaluated for two *real, non-isotropic* pair densities of the 74-determinant reference against brute-force Monte-Carlo: a signed p-like core×bond transition density on the leaves (bridge = the bond density) and a strongly directional bond density on the leaves (bridge = the core). In both the reduction reproduces the Monte-Carlo value—for the signed leaf a six-dimensional route that isolates the Coulomb bridge reaches $\sigma_{\mathrm{MC}}/|I| = 5 \times 10^{-4}$ and gives $|I_{\mathrm{reduction}} - I_{\mathrm{MC}}| = 1.1 \times 10^{-8}$ against a tolerance 1.7×10^{-8} (the twelve-dimensional estimator floors at 1.5×10^{-3} on the signed density and agrees at 0.3σ); for the directional leaf the unbiased twelve-dimensional integral confirms the reduction to 1.8×10^{-4}. Both dressed leaves carry Neumann channels beyond the $2\,l_{\mathrm{bridge}}$ (isotropic-leaf) bound—an $l = 4$ contribution at the level of 10^{-3} of the integral—so the general termination bound $2(l_{\mathrm{bridge}} + l_{\mathrm{leaf}})$ is the operative one, as the audit noted. The production f-kernel dressing itself contributes only $\sim 10^{-6}$ relative to these integrals, so it is adequate for sub-mHa energies without the exact-integration treatment the Coulomb kernel required. This upgrades the four-body bridge reduction from validated-on-model-densities to validated-on-real, non-isotropic densities under a non-block-factorized reference: the reference-generality reading of the audit paragraph is confirmed at the integral level. The assembled many-determinant *energy* followed: all seven energy enumerators were generalized to contractions over the coefficient tensor and run on a 74-determinant reference, giving a deterministic explicit-r_{12} LiH energy of -8.0372 Ha that an independent variational Monte-Carlo of the same trial function confirms to $\lesssim 0.3$ mHa (Paper 19, the 2026-09-23 marriage note). Backing: fast guard `tests/test_lih_r12ci_bridge.py` (the reduction versus its frozen Monte-Carlo value, with built-in fire tests that reject the isotropic-leaf shortcut and the $L \leq 2$ truncation on the p-like leaf).

[**VALIDATED 2026-09-26**] One caveat on the reference-generality reading concerns the triangle. When the shared vertex's pair density carries azimuthal circulation, which happens on a π-containing reference, the triangle reduction sums over mixed mode pairs (A, B) of its two correlation edges. Each nonzero mode on either edge carries a factor $\frac{1}{2}$, so the weight is $2\pi \cdot (\frac{1}{2})^{[A \geq 1]} (\frac{1}{2})^{[B \geq 1]}$. The first π-capable implementation applied the factor on one edge only. That gave $\pi/2$ instead of π for $A \geq 1$, $B = 0$ and shifted LiH energies by at most 0.18 mHa. An independent Monte-Carlo and a symmetry-orbit test caught and fixed it (Paper 19, the 2026-09-27 π-reference note). The two-orbital results of this section are unaffected, because their shared-vertex density is azimuthally uniform and the mixed pairs never occur. Guard: `tests/test_lih_gvee_triangle_symmetry.py`.

Backing: tests `tests/test_paper12_r12.py` (the H_2 engine) and `tests/test_lih_r12ci.py` (the two-center LiH engine), and the reduction of Eq. (33) above; the two-center bridge, the triangle reduction, and the full analytic 2×2 energy are validated against variational Monte-Carlo and chronicled in `CHANGELOG.md` (v5.15.10–v5.15.15). The proof-of-concept engine has been migrated from the sprint tree into the deterministic, resolution-of-identity–free `geovac.lih_r12ci` package (an `energy(geminal)` API reproducing both the -7.942 and -7.917 energies); it remains a proof-of-concept (ionic single-ζ reference), not a production chemistry path.

VIII. IMPLICATIONS FOR THE GEOVAC PROGRAM

A. The natural geometry hierarchy

Papers 0, 1, and 7 established S^3 as the natural geometry for one-center, one-electron systems. Paper 11 established prolate spheroids as the natural geometry for two-center, one-electron systems. This paper completes the analysis for two-center, two-electron systems and identifies the hierarchy of natural geometries:

1. **One-center, one-electron** (H, He^+): Three-

sphere S^3 via Fock projection. The Schrödinger equation becomes a Laplace–Beltrami eigenvalue problem. Exact on the labels (diagonal $SO(4)$ Casimir $n^2 - 1$); the graph Laplacian's $\lambda_{\max}$ is matched to the ground state by construction [1, 2].

2. **Two-center, one-electron** (H_2^+): Prolate spheroidal coordinates (ξ, η, φ). The Schrödinger equation separates into two coupled ODEs. Accuracy: machine precision (spectral Laguerre) [4].

3. **Two-center, two-electron** (H_2): Prolate spheroidal × prolate spheroidal with Neumann-algebraic V_{ee}. One-electron physics is exact; electron correlation is resolved to 92.4% in the σ sector, 99.1% once the azimuthal channels are restored (Sec. VII), and 99.81–99.97% after re-conditioning and the explicit-r_{12} extension (Secs. VII A, VII B). *This paper.*

Each level in this hierarchy addresses a qualitatively different physical singularity: the nuclear Coulomb singularity ($1/r$), the two-center potential ($1/r_A + 1/r_B$), and the electron-electron Coulomb singularity ($1/r_{12}$). The GeoVac program identifies each singularity's natural coordinate system and works in its label set and channel structure, solving the separated equations within each channel (Paper 11's natural geometry principle; not a discretization of the Laplace–Beltrami operator, which the S^3 edge Laplacian is shown not to be in Paper 7).

[WITHDRAWN] A fourth entry appeared here in earlier versions—"three-body coalescence: hyperspherical coordinates for the interelectron cusp, identified but not yet implemented"— presented as a consequence of this paper's residual. It is removed, because the residual was misdiagnosed (Sec. VI). This is a statement about what *this paper* licenses, not about hyperspherical coordinates, whose use for two-electron atoms rests on separate grounds in Paper 13 [5]. What the present result removes is the *evidence* that fourth entry rested on, not a verdict in its favour. Comparisons between the two geometries are meaningful only at matched $m_{\max}$, and at matched $m_{\max}$ neither this paper nor Paper 15 [6] claims an advantage over the other. The two published figures are not a matched pair: the 99.1% here is $(j_{\max}, l_{\max}) = (3, 3)$ at $|m| \leq 1$, while the 96.0% there is $l_{\max} = 6$ with a Schwartz cusp correction and a different solver class, so at matched $m_{\max}$ neither claims an advantage over the other. Beyond that matched comparison, re-basing the $|m| \leq 1$ span onto an orthogonal-polynomial family (Sec. VII A) takes this paper's H_2 to 99.81% of D_e, and the explicit-r_{12} extension (Sec. VII B) to 99.97% (0.053 mHa)—which settles constructively that the residual is electron correlation captured by ordinary integer powers of r_{12}, not an inability of these coordinates to represent the cusp. The defensible statement is the negative one: prolate spheroidal coordinates are not disqualified for H_2 by an inability to represent the cusp, because no such inability is in evidence.

B. Quantum chemistry without six-dimensional quadrature

The Neumann algebraic V_{ee} realizes a form of quantum chemistry in which the two-electron integrals are reduced entirely to recurrence relations and closed forms on quantum number labels—with the log-singular second-kind moment B_l now closed-form (Sec. III D), *no* numerical integration remains in either the σ or the $|m| \geq 1$ sector. Combined with the analytical kinetic energy (integration by parts) and nuclear attraction (simple polynomial moments), the *entire* Hamiltonian matrix is constructed without any quadrature at all.

This has conceptual significance beyond computational efficiency: it demonstrates that the physics of electron-electron repulsion in prolate spheroidal coordinates is fully encoded in the algebraic structure of the Legendre functions P_l and Q_l, not in the spatial geometry of a quadrature grid. The Neumann expansion is not an approximation; it is the exact representation of $1/r_{12}$ in the prolate spheroidal basis, truncated only by the angular momentum selection rule imposed by the basis itself.

C. Comparison with the S^3 approach

The S^3 label-set CI approach (Papers 0, 1, 7, and the FCI papers) constructs V_{ee} from Slater integrals (for s-pairs, the Fourier–Parseval F^0 form written on S^3 [3], [PRIOR ART]), which are computed analytically from hydrogen radial wavefunctions. That approach also avoids numerical integration of V_{ee}. On the physical Hamiltonian its He fixed-k ($k = Z$) hydrogenic CI reaches 1.88% at $n_{\max} = 7$ (Paper 13); the smaller atomic error formerly quoted for it came from a non-physical graph-hop one-body model (retired 2026-09-25), not from the physical Hamiltonian.

The prolate spheroidal approach with Neumann V_{ee} extends this philosophy to molecules: the natural coordinate system for each class of problems determines the algebraic structure of the two-electron integrals. On S^3, the algebraic structure is the Slater expansion in radial quantum numbers. On the prolate spheroid, it is the Neumann expansion in Legendre orders. Both are special cases of the general principle: *in the natural coordinate system, the Coulomb kernel has a known multipole expansion that converts integrals into algebraic sums.*

IX. CONCLUSION

We have demonstrated that the Neumann expansion of $1/r_{12}$ in prolate spheroidal coordinates eliminates the electron-electron integration bottleneck in the two-electron prolate spheroidal CI for H_2. The key results are:

1. **92.4% of exact** D_e with fully quadrature-free al-

gebraic V_{ee} (the log-singular B_l moment closed-form, Sec. III D) and zero fitted parameters.

2. **12–20 percentage point improvement** over numerical V_{ee} at every basis size from 1 to 72 functions.

3. **Numerical V_{ee} saturates at ∼80%** regardless of basis size, confirming grid truncation error as the dominant error source.

4. **27 basis functions is the sweet spot**: 92.2% in 705 s versus 92.4% in 5271 s at 72 functions.

5. **The 7.6% gap** is a basis-space limit, not an integration error—and specifically a *configuration* limit: the φ-independent basis spans only $m_1 = m_2 = 0$, while a $^1\Sigma_g^+$ state constrains only $M = m_1 + m_2$. The π^2 and δ^2 configurations it excludes carry the angular correlation.

6. **99.1% of exact** D_e (stability envelope 99.0–99.1%) on restoring the azimuthal channels at $|m| \leq 1$ in the same basis with the same Neumann kernel (Sec. VII), a gain of 11.6 mHa, against 0.34 mHa from near-tripling the σ basis. The σ-only value reproduces independently in a Gaussian basis to 0.2 mHa, and the extended value to within the stated stability envelope.

7. 99.81% **of exact** D_e (0.32 mHa, chemical accuracy) once that 99.1% monomial basis is re-conditioned and the shared exponent is taken to its variational optimum $\alpha = 1.40$ (99.77% at the fixed $\alpha = 1.0$ of Table IV): its ξ^j radial factor is a Hankel moment problem (the 99.1% is a conditioning wall, not a ceiling), and re-basing the same span onto orthogonal polynomials is an exact change of basis that lets the energy climb monotonically and variationally to chemical accuracy at $(5, 5) + \delta$ (Sec. VII A). What the residual is *not* is more azimuthal channels; radial completeness and the e–e cusp remain candidates that the knob elimination does not separate.

[WITHDRAWN] Earlier versions of this paper concluded that the residual was the electron-electron cusp demanding non-analytic $r_{12}^{1/2}$ and $r_{12} \ln r_{12}$ terms, and read that as identifying hyperspherical coordinates as the next natural geometry. Both are withdrawn: the calculations above contain no such term and reach chemical accuracy—0.053 mHa with the explicit-r_{12} extension—and an accurate grid-based prolate spheroidal treatment in the literature [7] independently shows these coordinates are not cusp-limited.

The result demonstrates a form of quantum chemistry where recurrence relations and closed forms on quantum number labels replace the six-dimensional quadrature for electron-electron repulsion entirely—including the log-singular B_l moment, whose closed form (Sec. III D) carries the same construction to the associated Legendre functions of the $|m| \geq 1$ channels (Sec. VII), so no numerical integration remains. Just as Paper 7 showed that atomic energies are encoded in the topology of S^3, this paper shows that molecular electron-electron repulsion is encoded in the algebraic structure of the Legendre functions P_l, Q_l and their associated forms P_l^m, Q_l^m on the prolate spheroid. The physics is in the algebra, not in the grid.

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Appendix A: Numerical Verification of Auxiliary Integrals

All auxiliary integrals were validated against independent numerical quadrature. The test suite (`tests/test_neumann_vee.py`, 32 tests) verifies:

- $C_l(q)$: Base cases (C_0 even/odd, C_1 odd), selection rule, and known analytical values. Agreement to machine precision ($< 10^{-14}$).
- $A_n(\alpha)$: Comparison with `scipy.integrate.quad`. Agreement to $< 10^{-12}$.
- $A_l(p, \alpha)$: Comparison with numerical quadrature of $\xi^p e^{-\alpha\xi} P_l(\xi)$. Agreement to $< 10^{-10}$.
- $B_l(p, \alpha)$: Comparison with independent adaptive quadrature. Agreement to $< 10^{-10}$.
- $X_l(p_1, p_2, \alpha)$: Comparison with 2D numerical quadrature. Agreement to $< 10^{-8}$. Symmetry $X_l(p_1, p_2) = X_l(p_2, p_1)$ verified to $< 10^{-14}$.
- V_{ee} **matrix**: Symmetry, positive diagonal, and comparison with numerical elliptic integral quadrature. Neumann vs. numerical agreement to $<$ 0.01 Ha per element (the discrepancy is the grid error in the numerical values).
- **Full Hamiltonian**: Variational principle (energy above exact), bound state, and consistency between Python and Numba code paths ($< 10^{-8}$ Ha).

Appendix B: Comparison with r_{12}-Dependent Basis Functions

We tested basis functions with explicit r_{12}^p dependence ($p = 1, 2$), following the James–Coolidge approach [13]. The r_{12} factor is expressed in prolate spheroidal coordinates as a function of $(\xi_1, \eta_1, \xi_2, \eta_2, \varphi_1 - \varphi_2)$, making the integrals five-dimensional after azimuthal averaging.

TABLE V. Comparison of r_{12}-dependent basis (numerical V_{ee}) vs. $p = 0$ basis (Neumann V_{ee}). All results at $R = 1.4011$ bohr.

Method	N	D_e%	t (s)
r_{12}^1, num., 60×40 grid	6	85.7	447
r_{12}^{0-2}, num., 20×14 grid	18	86.8	970
p=0, Neumann, 30×20 grid	27	92.2	705
p=0, Neumann, 30×20 grid	72	92.4	5271

Table V compares the best r_{12}-dependent results with the $p = 0$ Neumann results.

The Neumann $p = 0$ result at 92.2% exceeds the best r_{12}-dependent numerical result (86.8%) by 5.4 percentage points, confirming that exact V_{ee} evaluation is more important than r_{12}-dependent basis functions when the V_{ee} integrals are computed numerically. The combination of r_{12}-dependent basis functions with Neumann-type algebraic V_{ee} is a natural extension left for future work.

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